1. Linear Equations in Two Variables

1. Solve by elimination: $5x - 3y = 7$ and $3x + 2y = 1$.

2. Solve by substitution: $x + 2y = 4$ and $3x - y = 5$.

3. Find the point of intersection of lines $2x + 3y = 6$ and $4x - 3y = 12$.

4. Form a pair of linear equations and solve: The sum of two numbers is 27 and their difference is 5.

5. A two-digit number is such that the sum of its digits is 11. If we interchange the digits, the new number is 9 less than twice the original number. Form the pair of linear equations and find the number.

6. The sum of the present ages of a father and his son is 50 years. Ten years ago, the father's age was three times the son's age. Find their present ages.

7. Draw the graph of $2x - 3y = 6$ and $x + y = 4$. Hence find the coordinates of their point of intersection and verify it algebraically.

8. For what value of k will the following pair have (i) no solution (ii) infinitely many solutions? $3x - 5y = 7$, $6x - 10y = k$.

9. A shop sells pencils at ₹5 each and pens at ₹12 each. In one day, the shopkeeper sold 40 items and collected ₹320. Form the pair of linear equations and find how many pencils and pens were sold.

10. Determine whether the lines represented by $4x + 6y = 12$ and $2x + 3y = 8$ are intersecting, parallel or coincident. Justify your answer.